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Revision History

Marketing PN	Rev.	Date	By	Description
X-NAVQ95-MAIN	X1	2026-05-07	JvE (CTO SI)	Initial Schematic Development
X-NAVQ95-MAIN	A	2026-06-11	JvE (CTO SI)	Schematic release
NAVQ95-MAIN	A1	2026-09-10	JvE (CTO SI)	Schematic release to production
NAVQ95-MAIN	A2	2026-10-12	JvE (CTO SI)	BOM change. Schematic release to production

NAVQ95-MAIN

Part of MR-NAVQ95-KIT (97275 rev. A)

NOTES:

1. Interrupted lines coded with the same letter or letter combinations are electrically connected.
2. Device type number is for reference only. The number varies with the manufacturer.
3. Greyed out components indicate Do Not Populate (DNP) during assembly.
4. Special signal usage:
   _B denotes an active-low signal.
   _P and _N denote a differential signal.
5. Interpret diagram in accordance with American National Standards Institute specifications, current revision, with the exception of logic block symbology.
6. All SILK display properties will be in this color
7. All schematic annotations will be in this color

System Innovation
High Tech Campus 60
5656 AG Eindhoven, The Netherlands

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Designer: Jari van Ewijk	Drawing Title: NAVQ95-MAIN		
Drawn by: Jari van Ewijk	Page Title: Cover		
Approved: Costin Oncfrei	Size C	Document Number SCH-97010 / PDF: SPF-97010	Rev A2
	Date: Tuesday, October 13, 2026	Sheet 1 of 25	

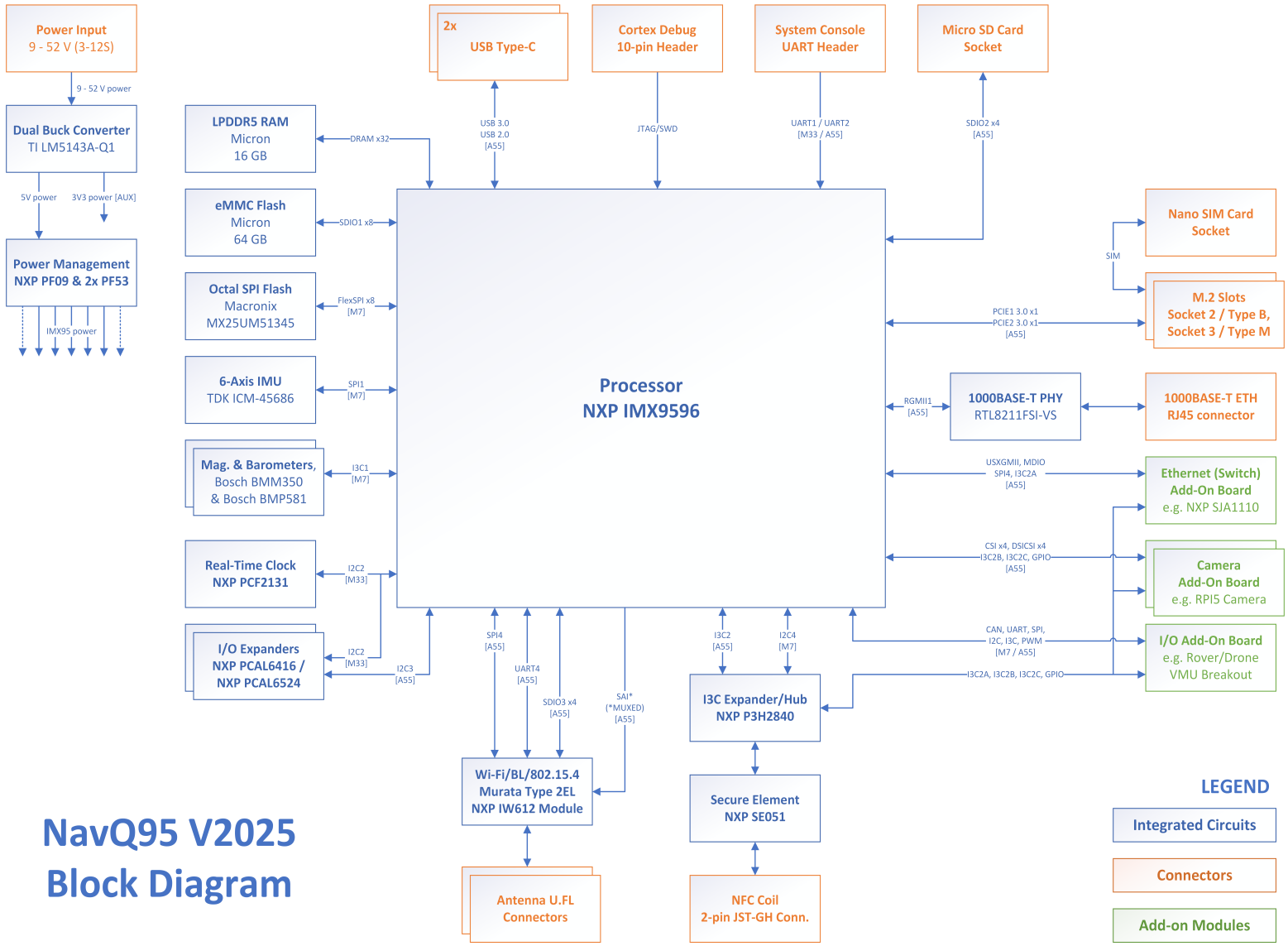
DNP Components

REF DES	ASSY_OPT	PAGE NAME
R124,R126,R269	DNP	05 - HV Buck Converter
C23,C29,C92,R65,R68,R73,R74, R145,R152,R156,R229,R230,R257	DNP	06 - LV PMICs
R112,R123,R137,R138,R139,R254	DNP	09 - MPU I/O
C181,D14,R250	DNP	10 - MPU Misc.
R292,R295,R296,R297,R298	DNP	11 - eMMC Flash
R141	DNP	12 - Octal SPI Flash
C81,R288,R290	DNP	13 - MicroSD Card
R52,R56,R60,R75,R78	DNP	17 - RTC & Clock Buffer
R378,R379	DNP	18 - Sensors
C427,R158,R159,R162,R164, R165,R166,R169,R361,R362, R364,R365	DNP	19 - Tri-Radio
R315,R317	DNP	20 - Gigabit Ethernet
R212	DNP	21 - M.2 Key M
R113,R114,R115,R266,R267, R268,R289	DNP	22 - M.2 Key B
R103,R176,R177,R178,R179, R180,R181	DNP	23 - I/O Expanders
R185	DNP	24 - Secure Element

Switch Default Positions

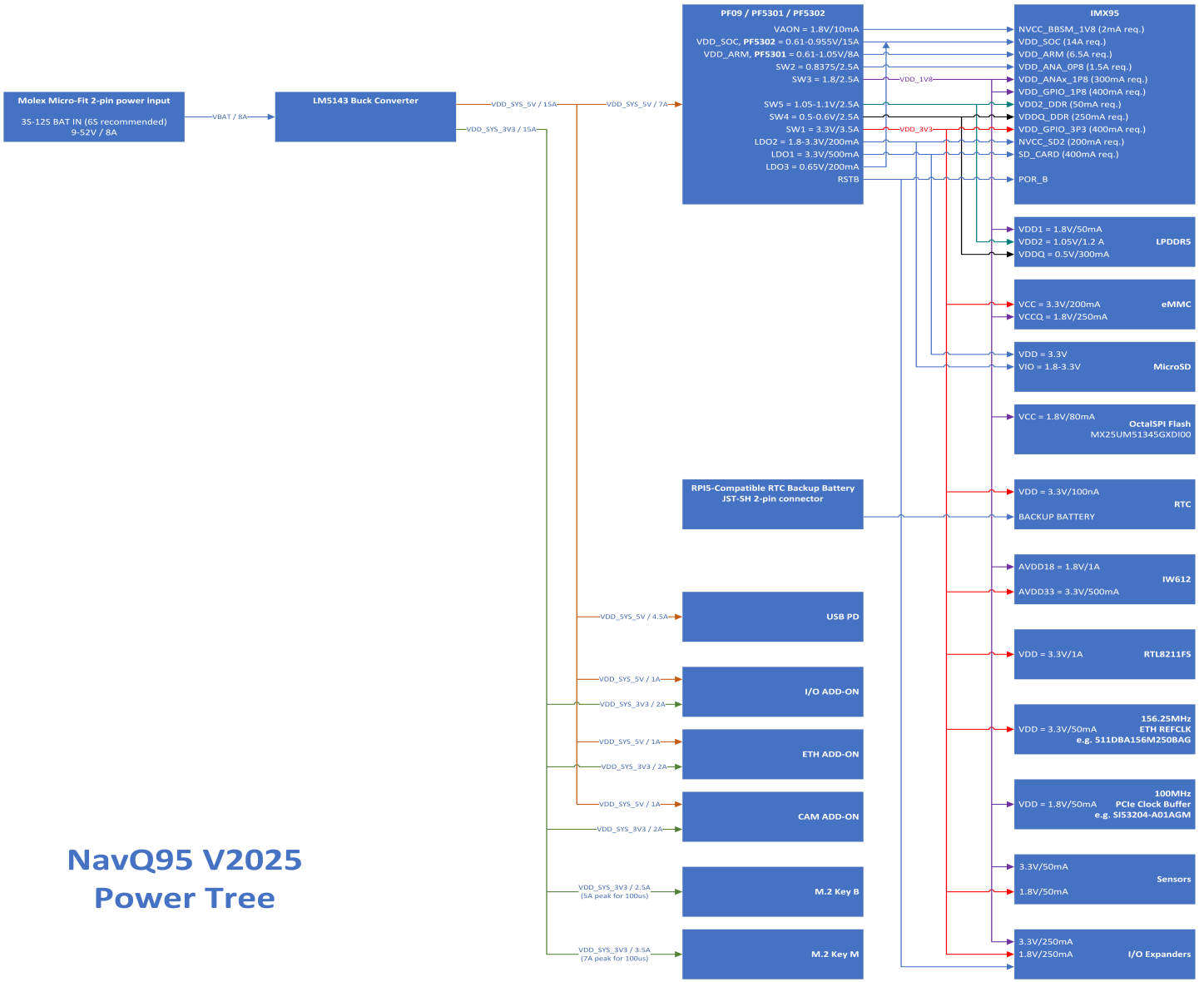
REF DES	SWITCH	PAGE NAME
SW2	2-4-5-8	14 - Buttons & Boot Mode

Classification: Public
Drawing Title: NAVQ95-MAIN
Page Title: Tables
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NavQ95 V2025 Block Diagram

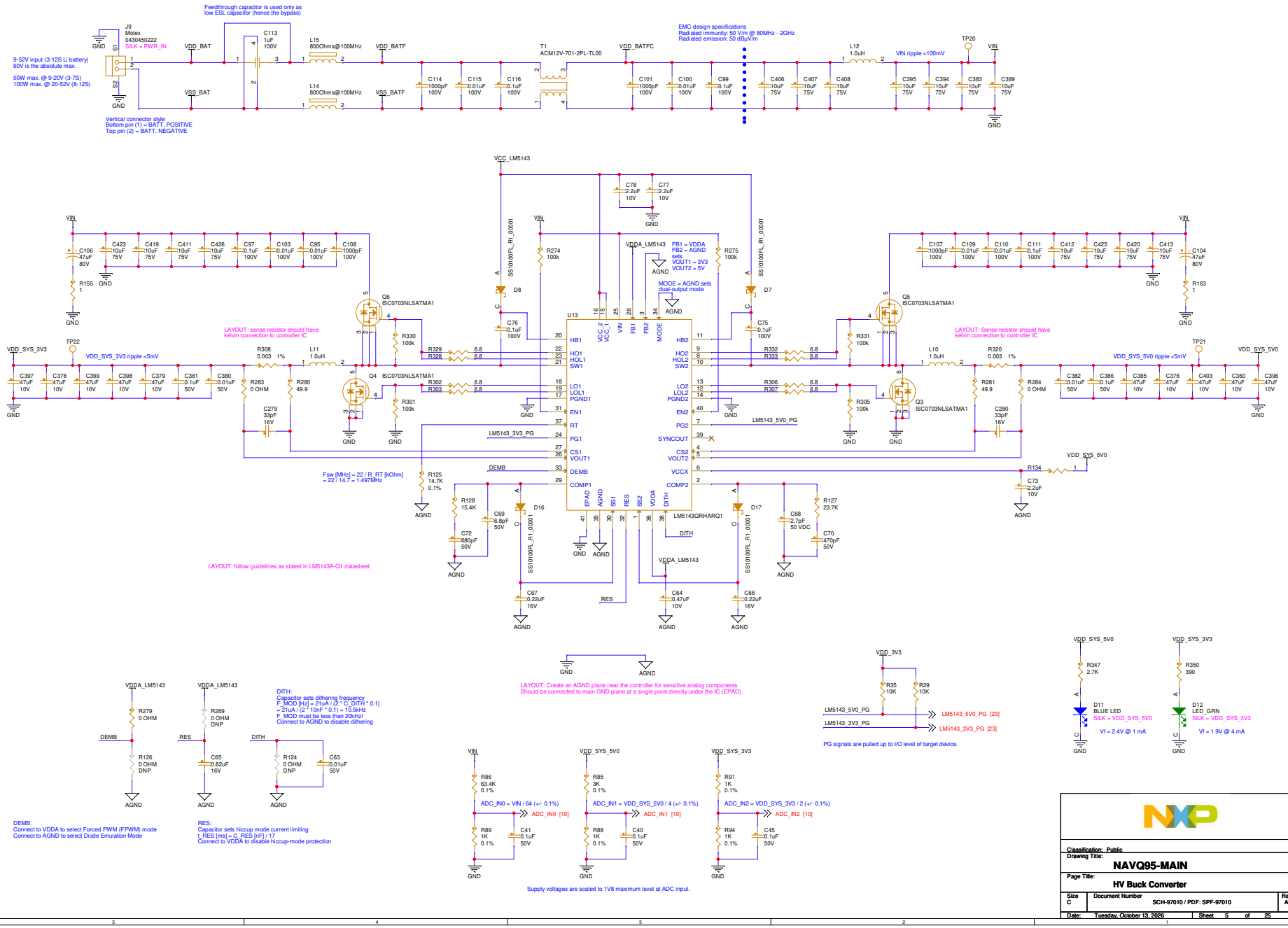
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Drawing Title:	NAVQ95-MAIN
Page Title:	Block Diagram
Size	C
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NavQ95 V2025
Power Tree

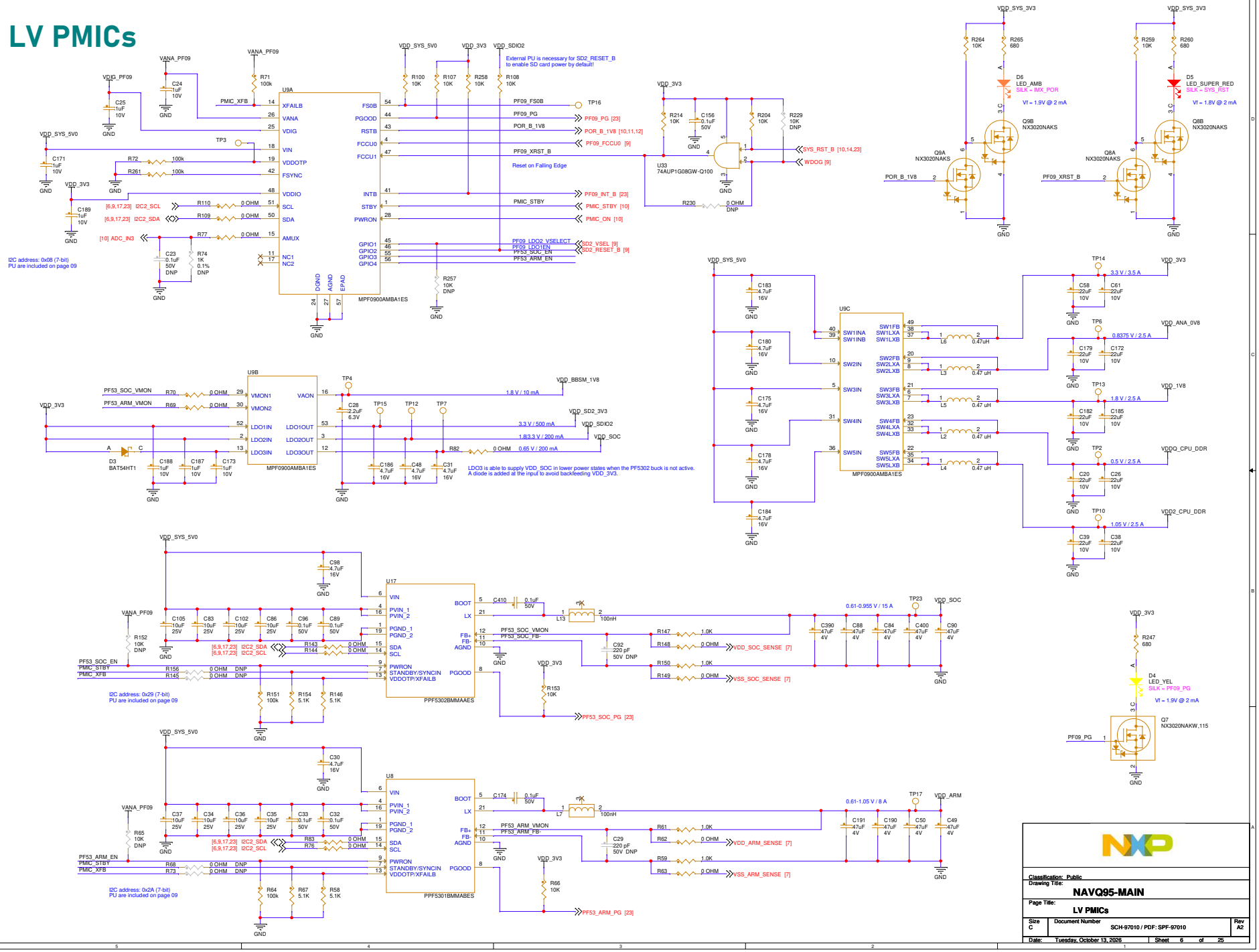
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Page Title: Power Tree
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HV Buck Converter



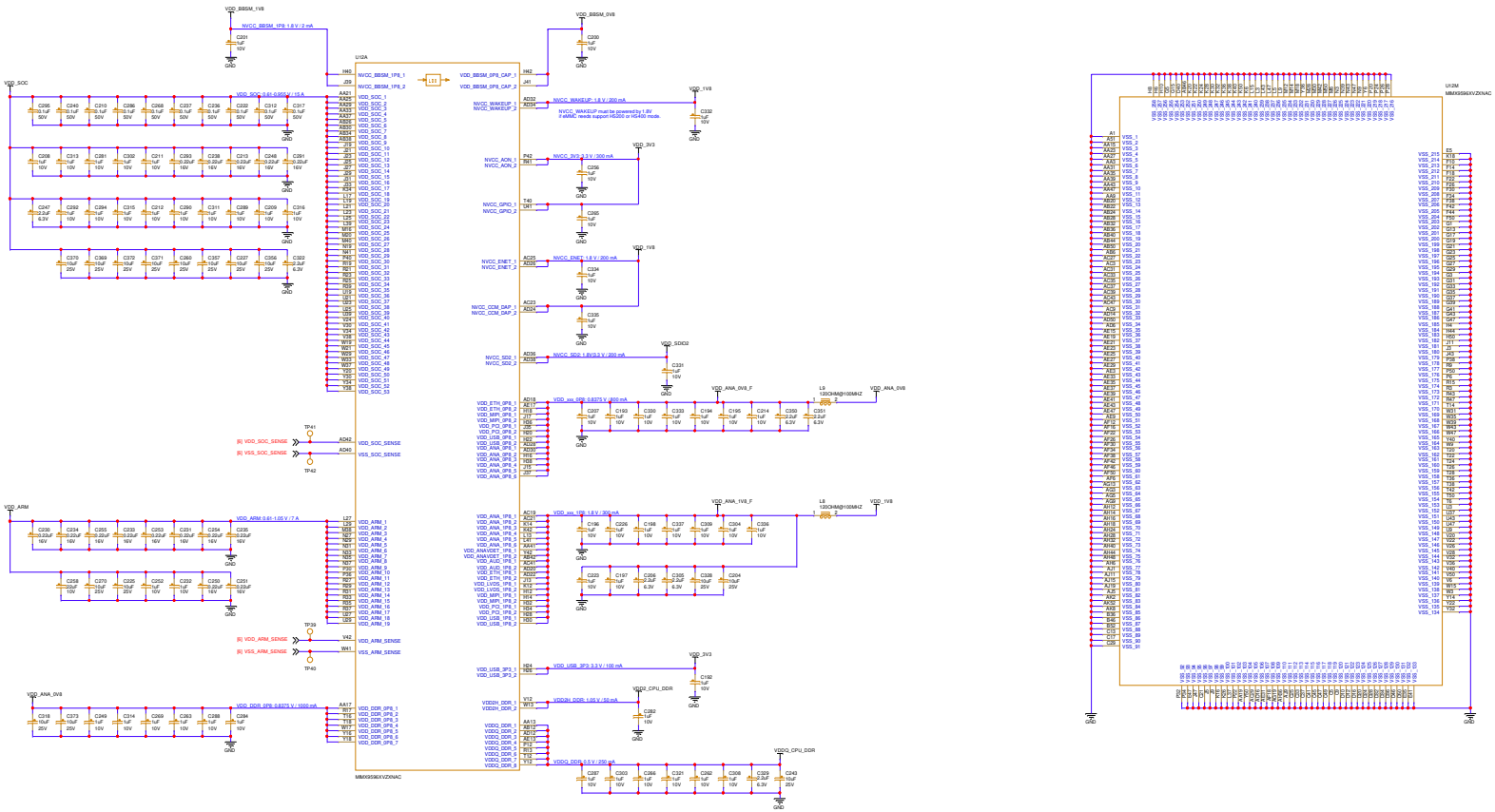
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Page Title:	HV Buck Converter
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LV PMICs



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MPU Power

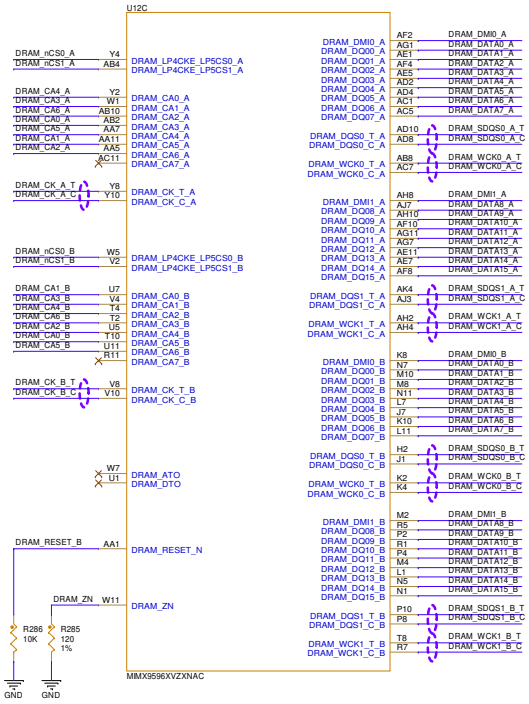


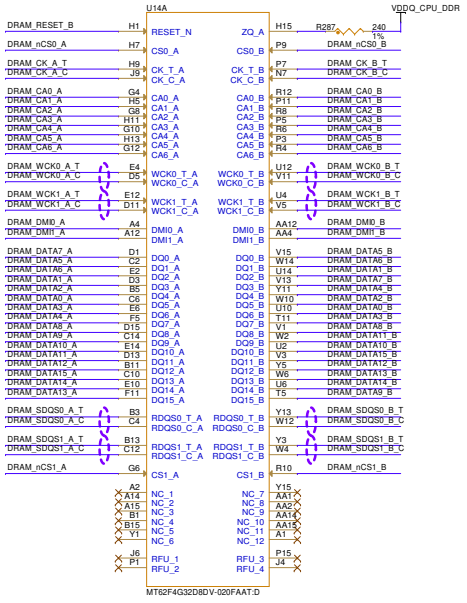
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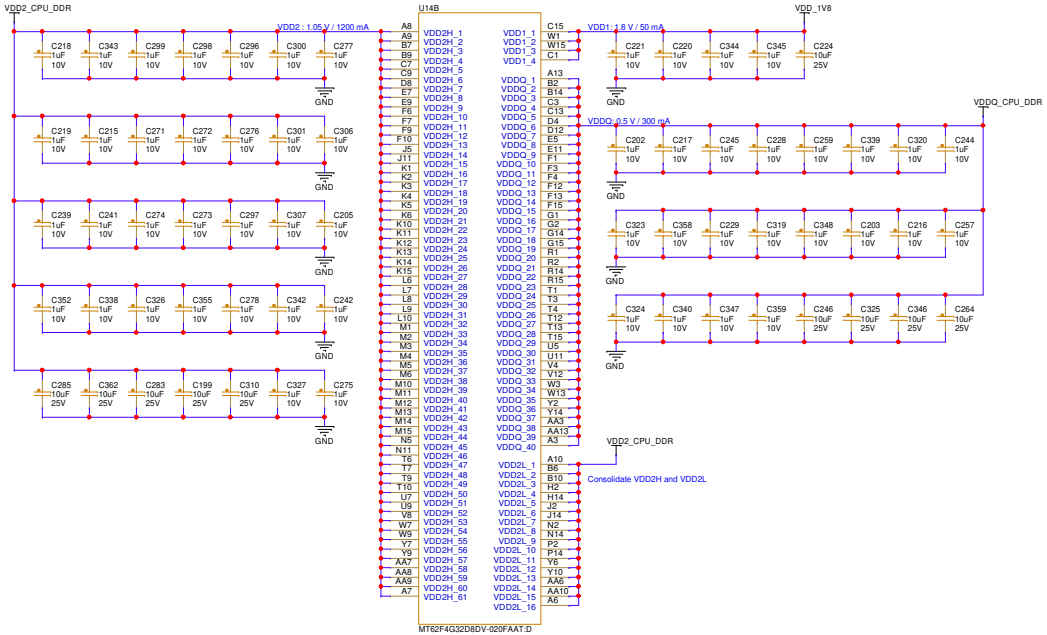
LPDDR5

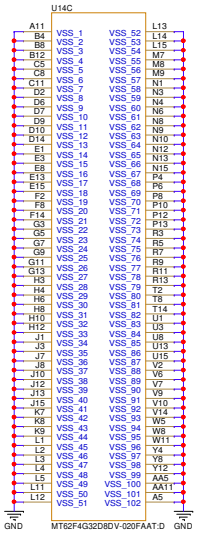
Note: net names are re-arranged (swizzled) to allow for optimized routing. CA net name follows memory device name, DQ net name follows CPU signal name.

DDR: 80 ohm









NXP

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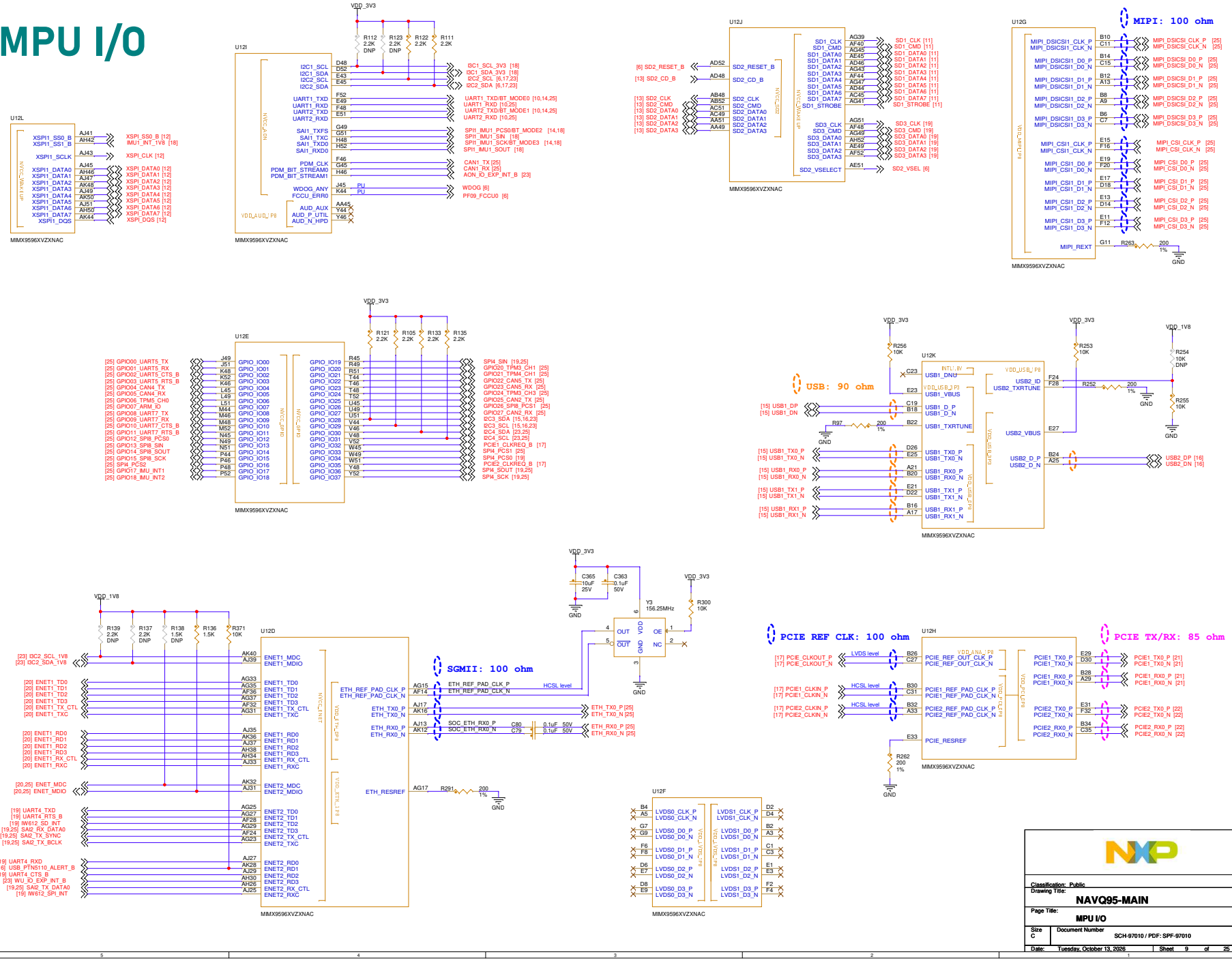
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Page Title: **LPDDR5**

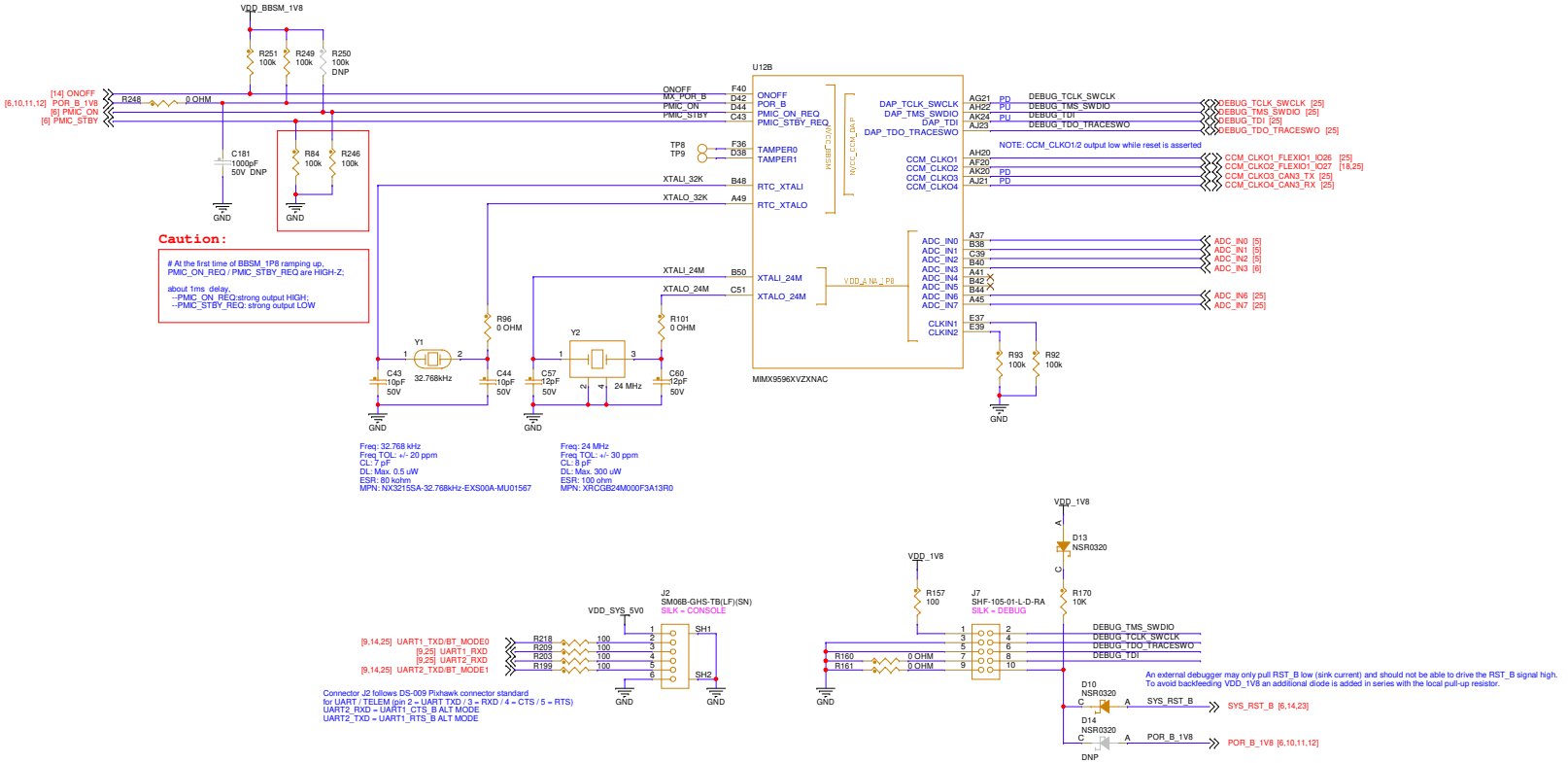
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MPU I/O


U12L
MIMX9596XVZXNAC
U12I
MIMX9596XVZXNAC
U12J
MIMX9596XVZXNAC
U12G
MIPI: 100 ohm
MIMX9596XVZXNAC
U12E
MIMX9596XVZXNAC
USB: 90 ohm
U12K
MIMX9596XVZXNAC
U12D
SGMII: 100 ohm
MIMX9596XVZXNAC
PCIE REF CLK: 100 ohm
U12H
PCIE TX/RX: 85 ohm
MIMX9596XVZXNAC
U12F
MIMX9596XVZXNAC
NXP
Classification: Public
Drawing Title:
NAVQ95-MAIN
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MPU Misc.



Classification: Public

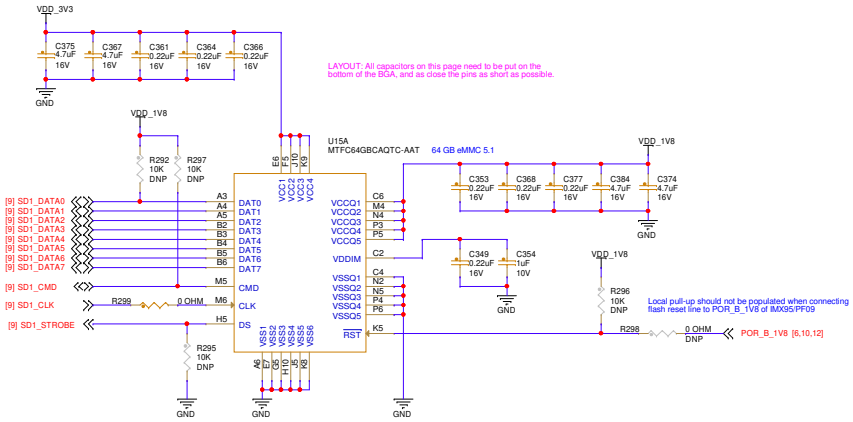
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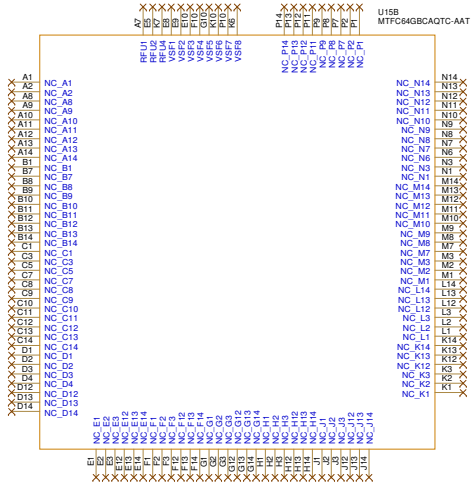
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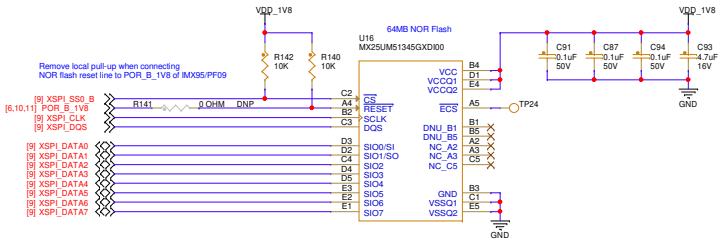
eMMC Flash





Classification: Public		
Drawing Title: NAVQ95-MAIN		
Page Title: eMMC Flash		
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Octal SPI Flash



Classification: Public

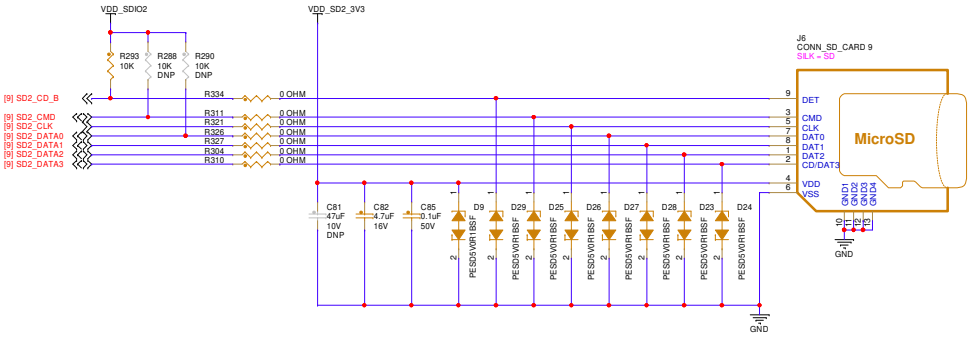
Drawing Title: NAVQ95-MAIN

Page Title: Octal SPI Flash

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MicroSD Card

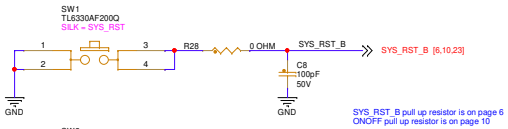


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Drawing Title: NAVQ95-MAIN		
Page Title: MicroSD Card		
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Buttons & Boot Mode

SYS RESET

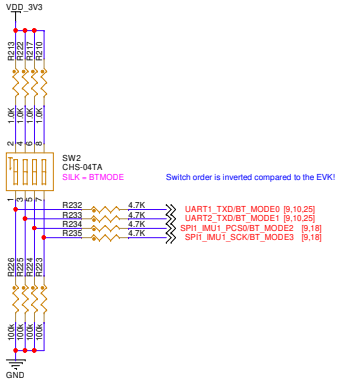
Short Press: Reset



ONOFF

Long Press 5 seconds: Turn OFF
Short Press: Turn ON





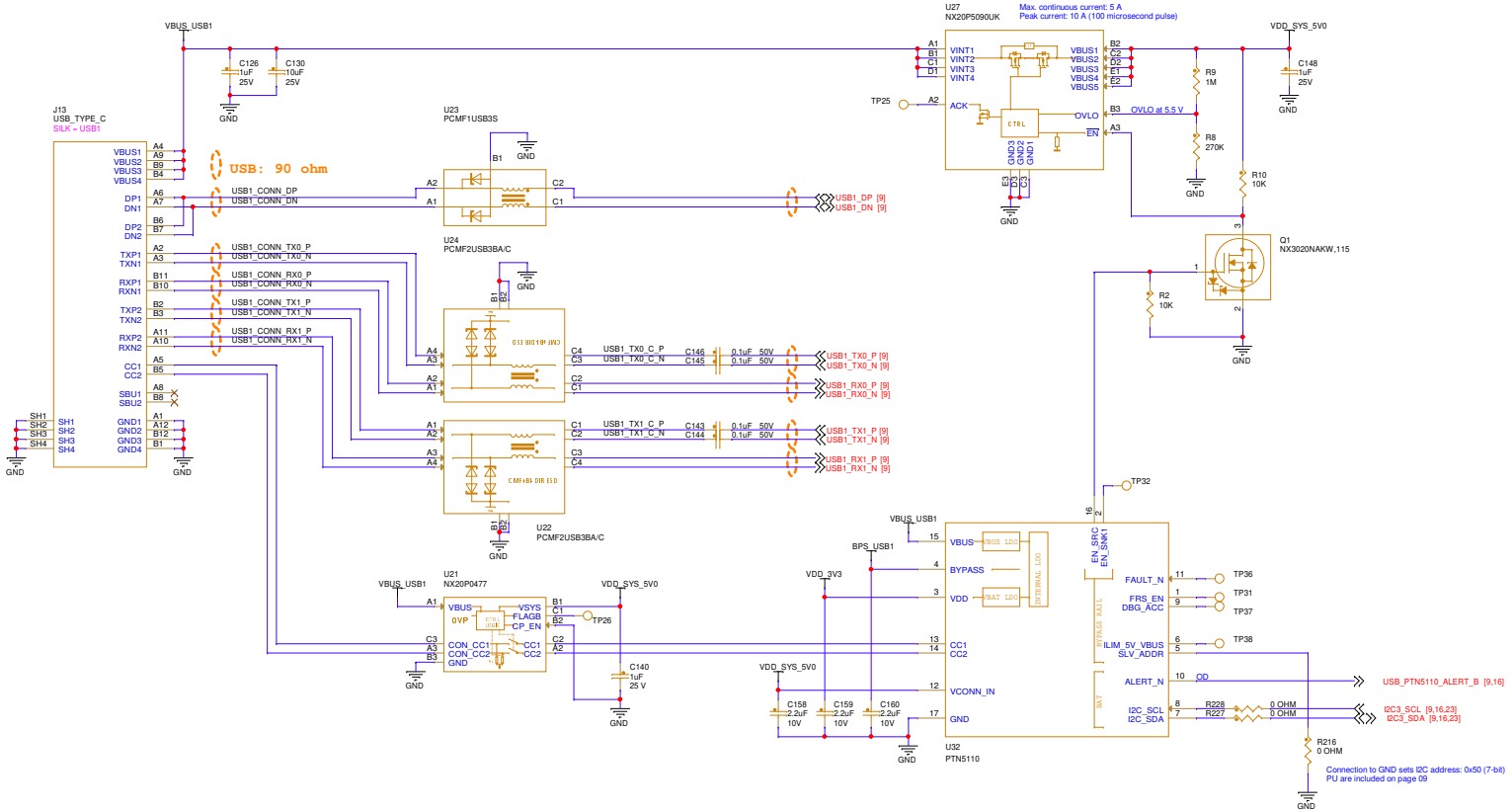
i.MX95 Boot Modes

BOOT_MODE[3:0]	SW2-1	SW2-2	SW2-3	SW2-4	BOOT DEVICE
1001	ON	OFF	OFF	ON	Serial Downloader on USB1 (J13)
1010	OFF	ON	OFF	ON	Boot from eMMC Flash (U15)
1011	ON	ON	OFF	ON	Boot from MicroSD Card (J6)
1100	OFF	OFF	ON	ON	Boot from Octal SPI Flash (U16)

Switch order is inverted compared to the EVK!
See IMX95RM for more details about different boot modes.

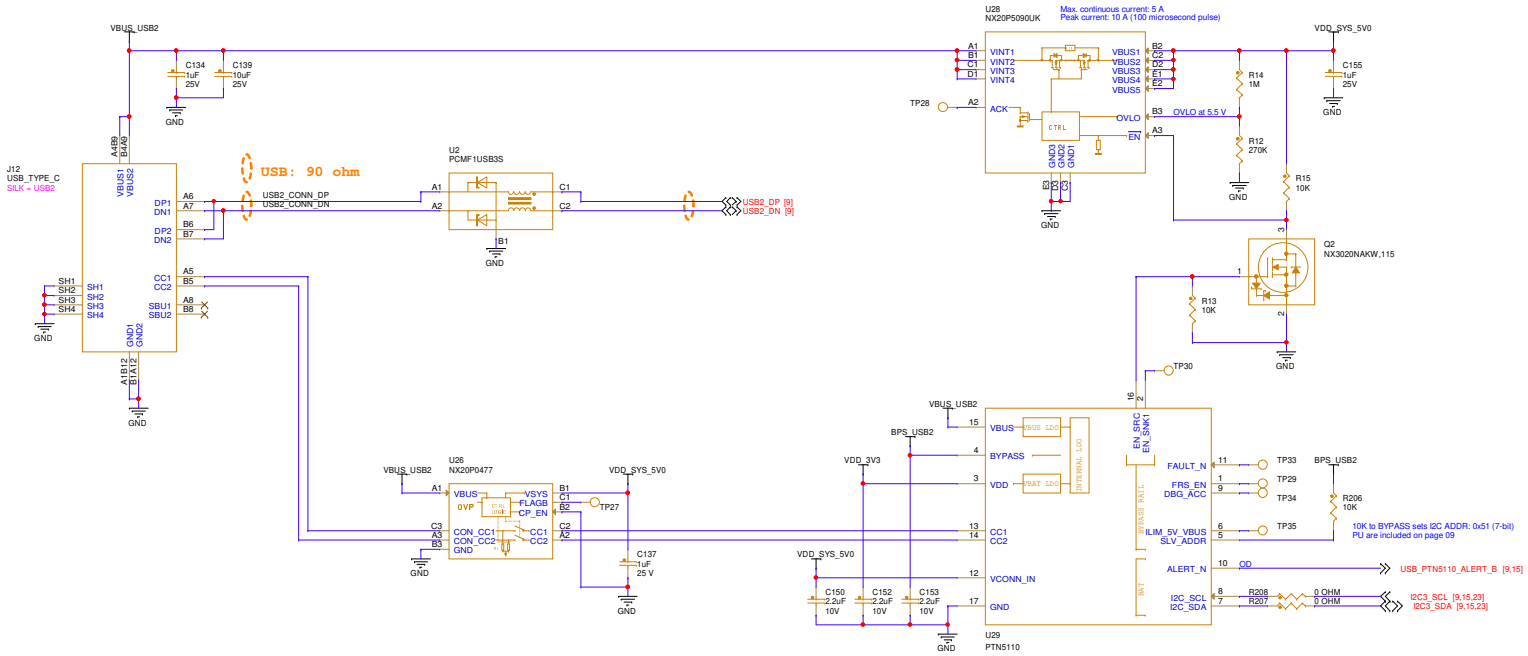
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Drawing Title: NAVQ95-MAIN
Page Title: Buttons & Boot Mode
Size: C
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USB1 3.0 Type C



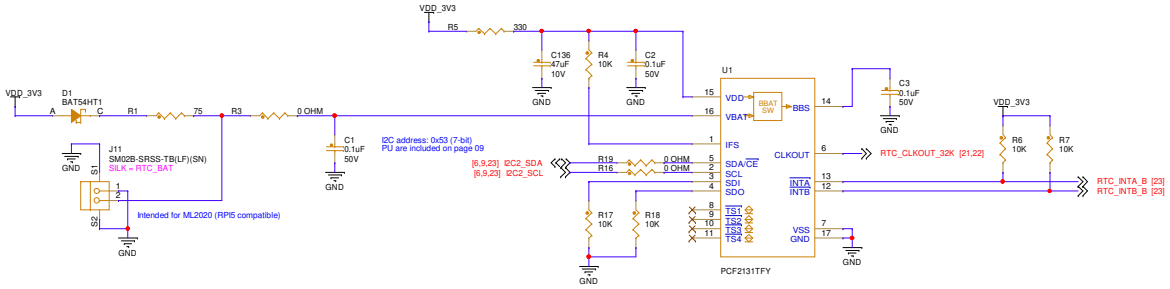
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Page Title: USB1 3.0 Type C		
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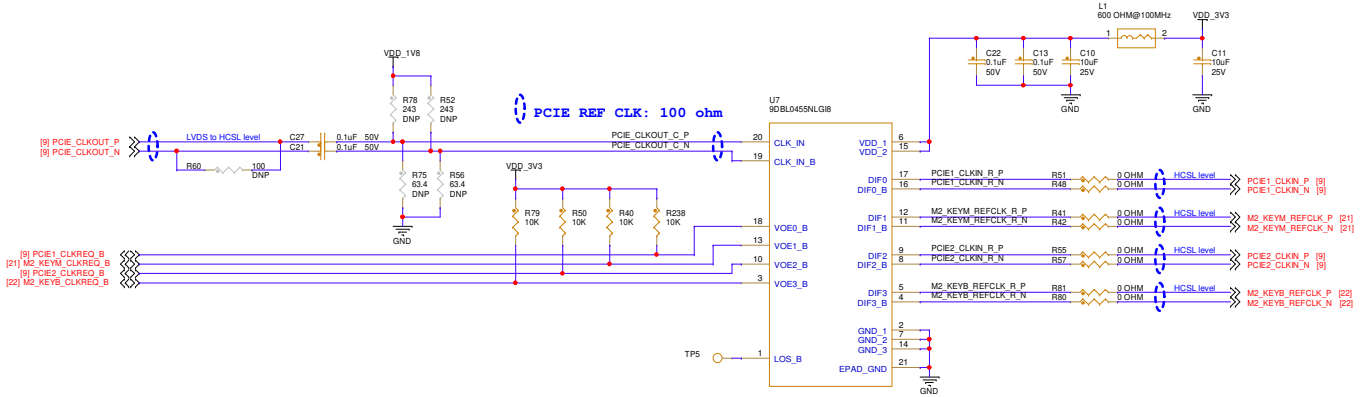
USB2 2.0 Type C



Classification: Public		
Drawing Title: NAVQ95-MAIN		
Page Title: USB2 2.0 Type C		
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RTC & Clock Buffer





Classification: Public

Drawing Title: **NAVQ95-MAIN**

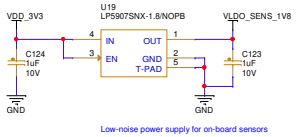
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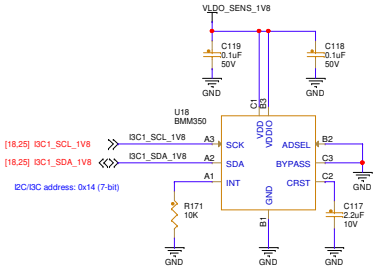
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Sensors

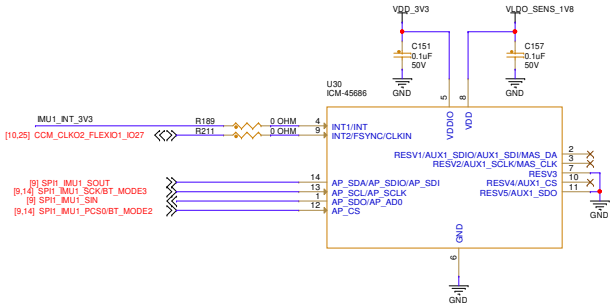
Low-Noise Regulator



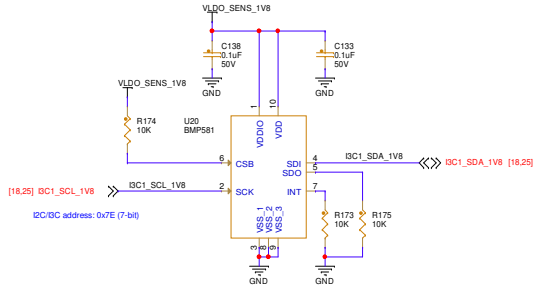
Magnetometer

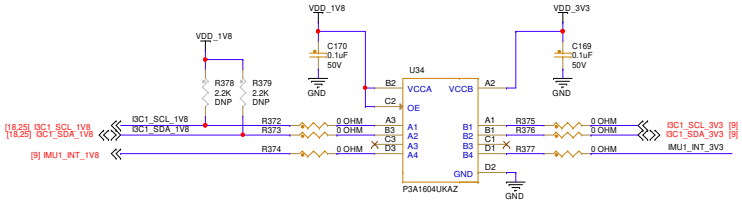


6-Axis Accelerometer & Gyroscope



Barometer



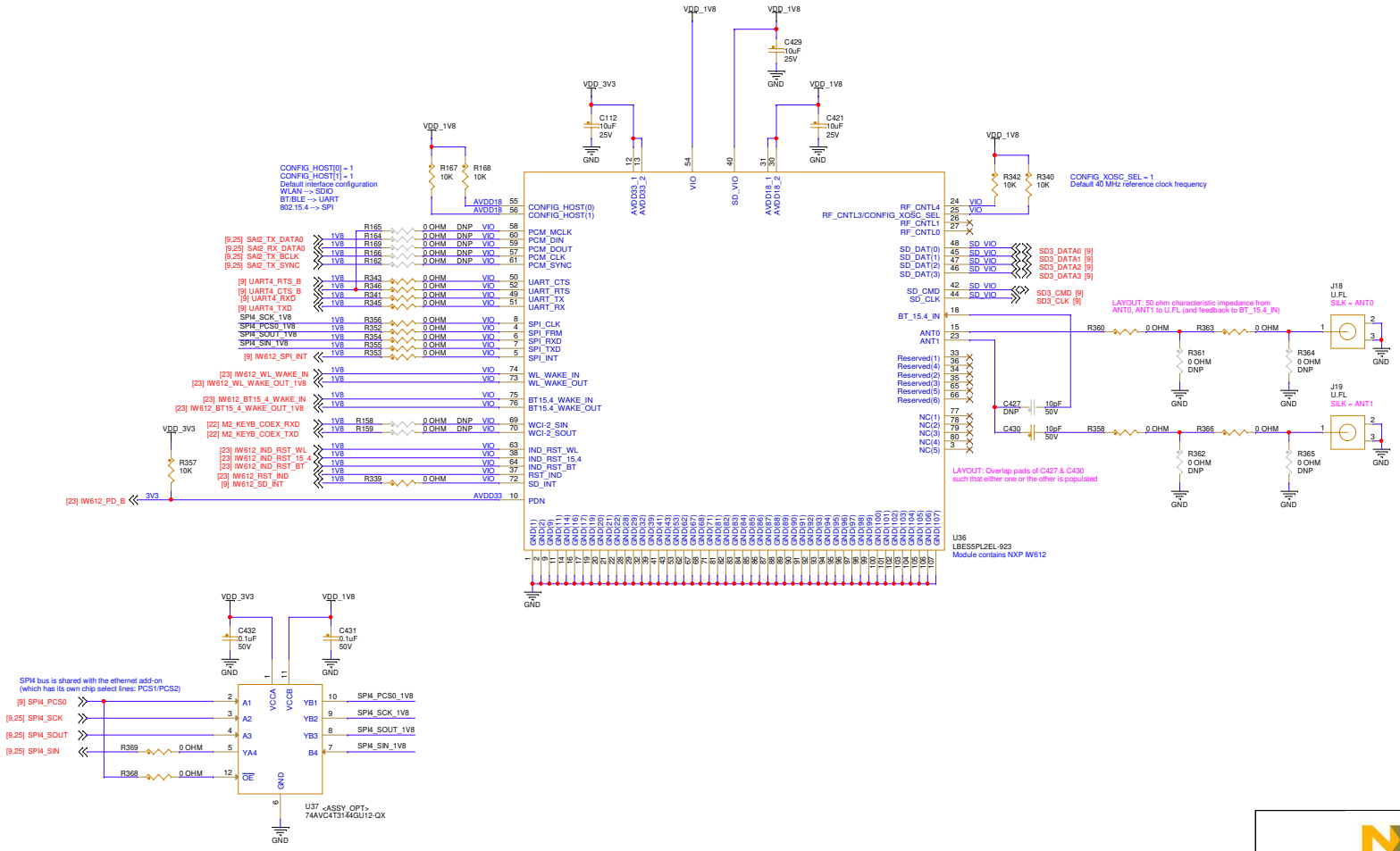


Barometer

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Drawing Title:	NAVQ95-MAIN
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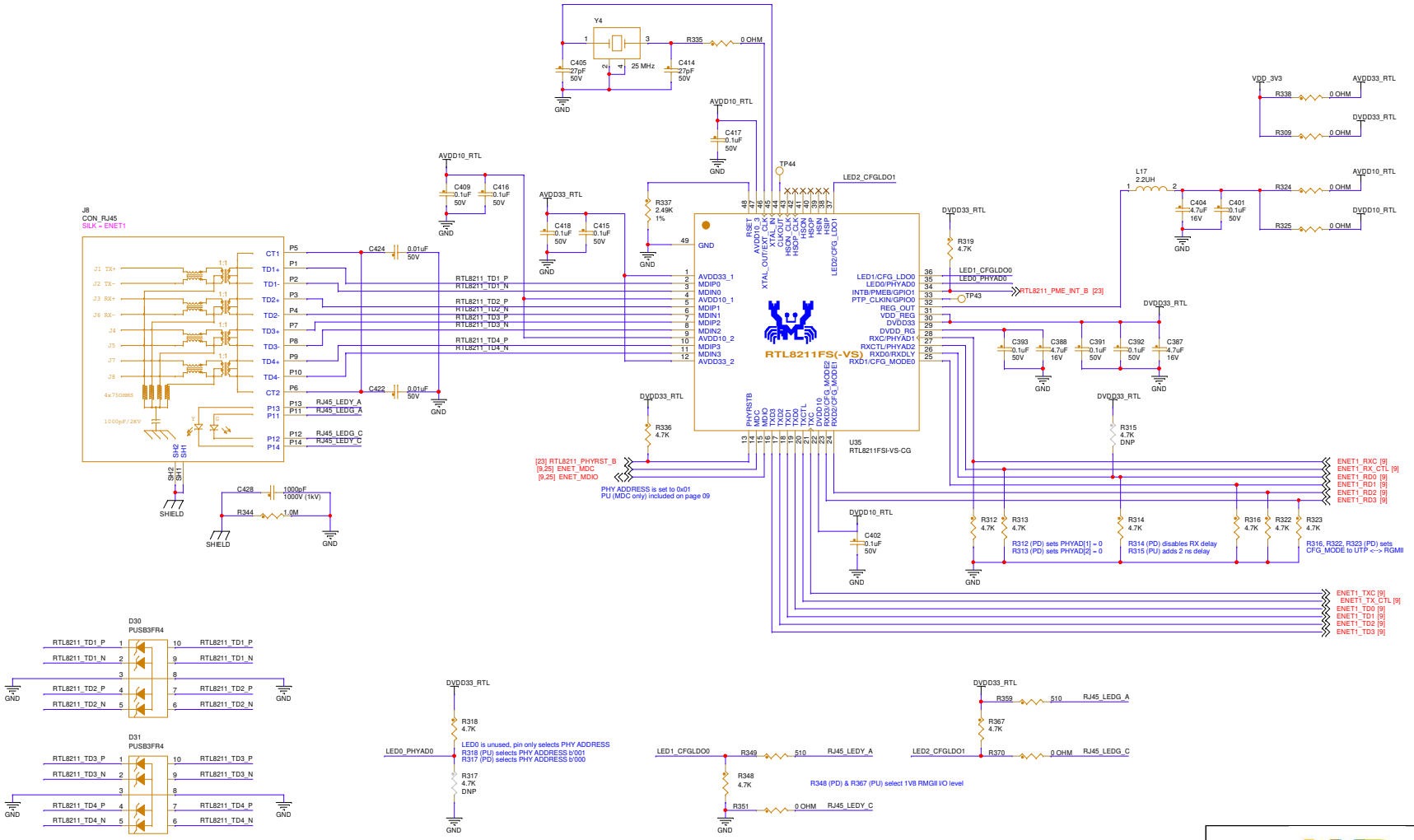
Tri-Radio

Wi-Fi 6 / Bluetooth 5.4 / IEEE 802.15.4



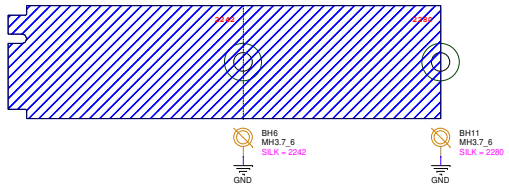
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Page Title: Tri-Radio		
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Gigabit Ethernet

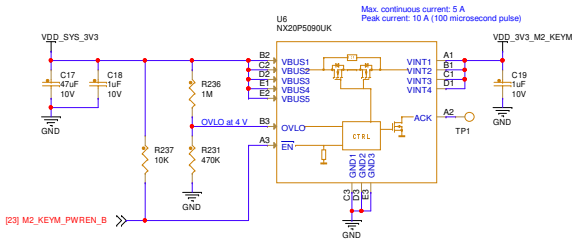


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Page Title: Gigabit Ethernet			
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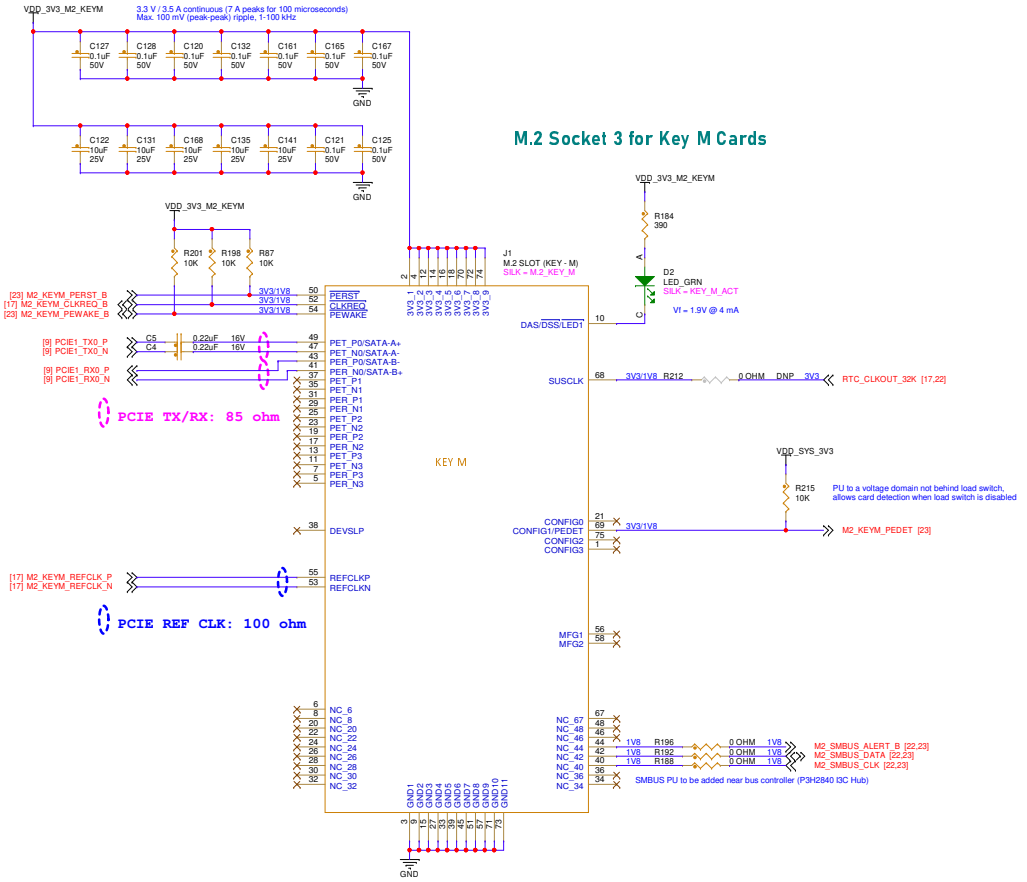
M.2 Key M



Load Switch

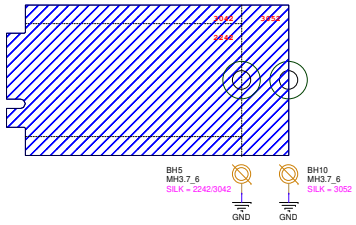


M.2 Socket 3 for Key M Cards

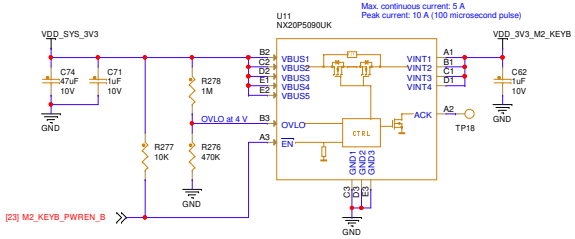


Classification: Public		
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Page Title: M.2 Key M		
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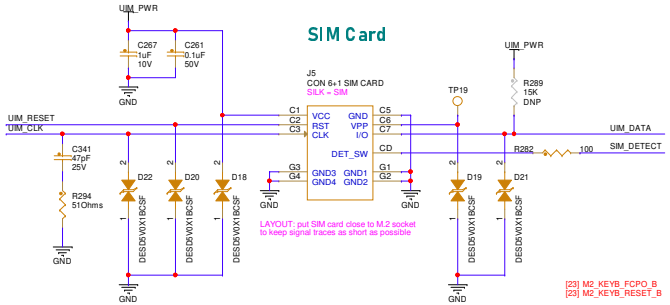
M.2 Key B



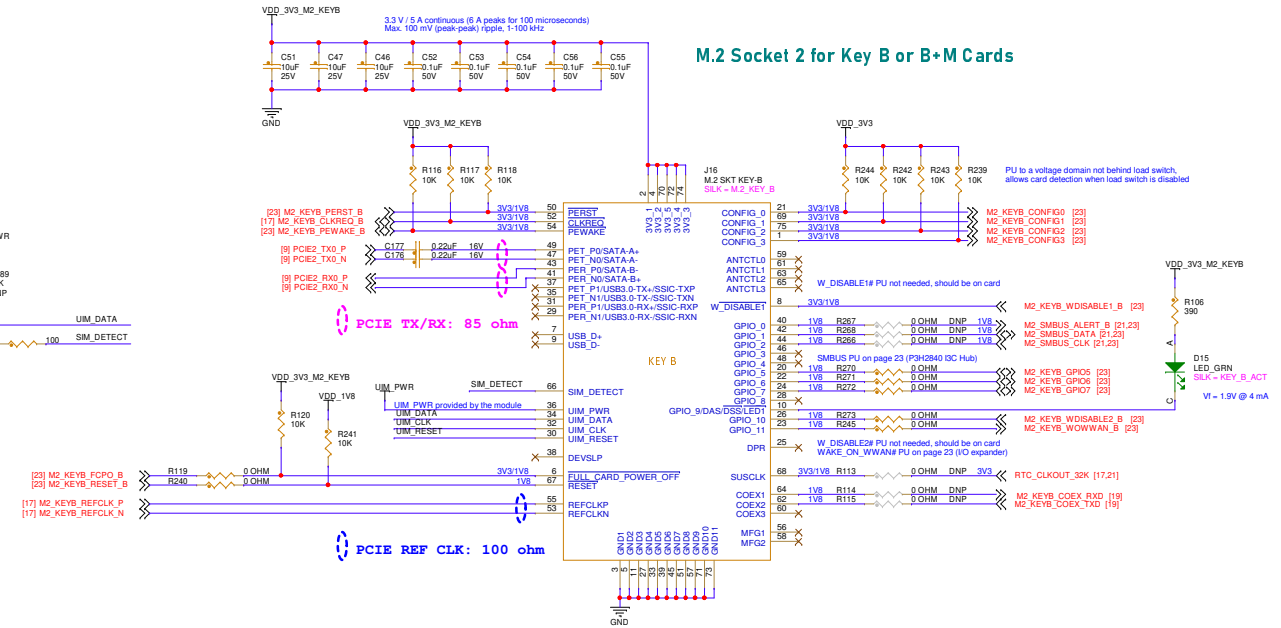
Load Switch



SIM Card

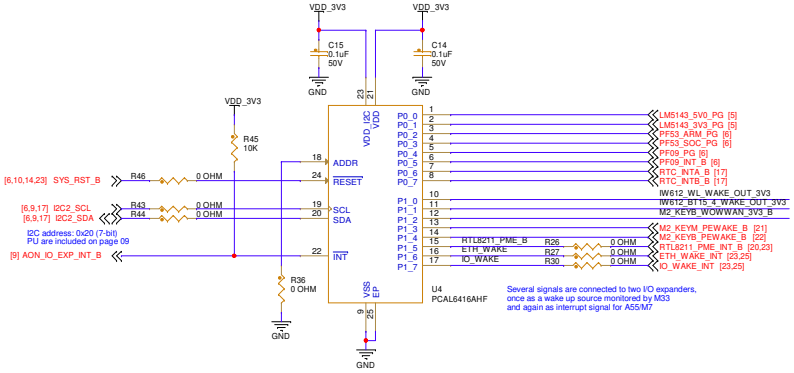


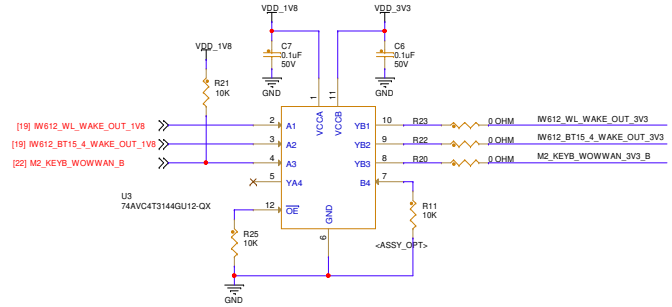
M.2 Socket 2 for Key B or B+M Cards

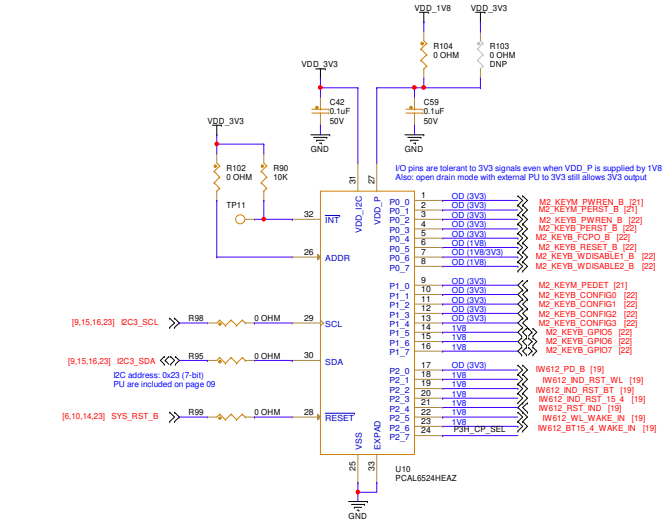


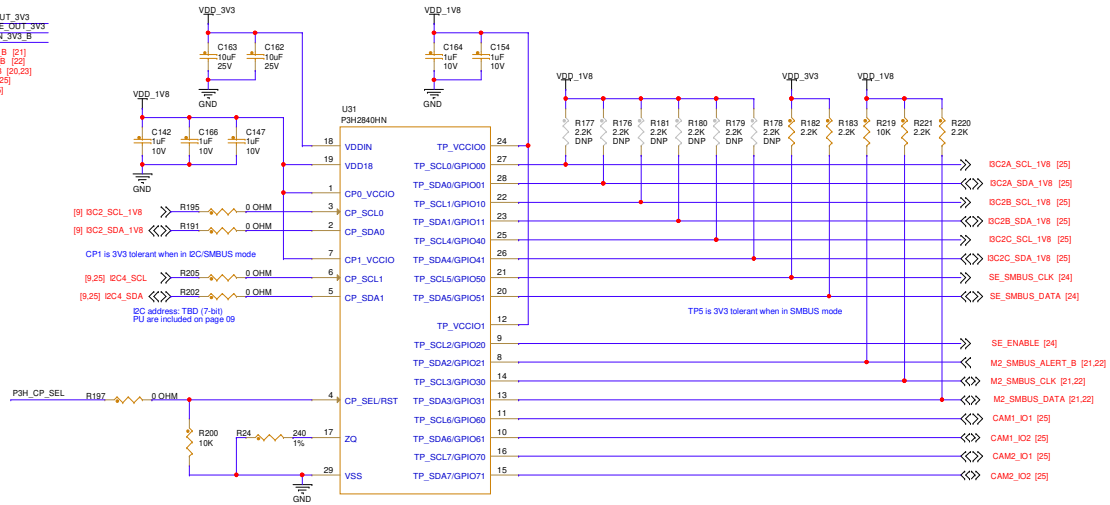
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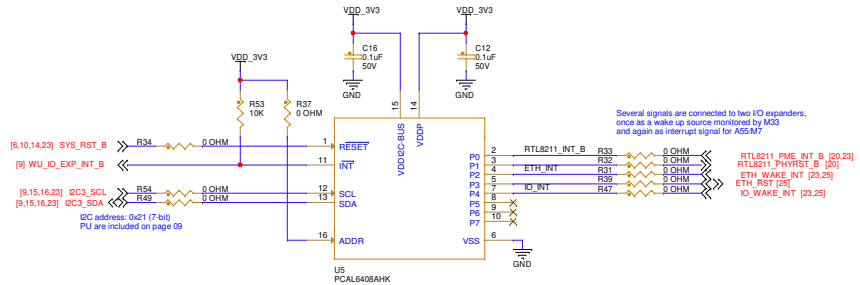
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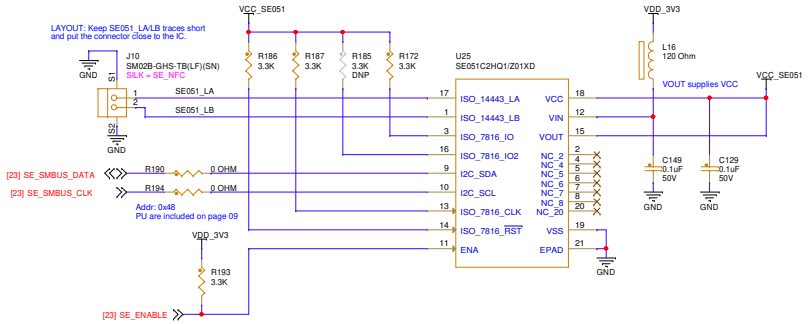
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Secure Element



Classification: Public

Drawing Title: **NAVQ95-MAIN**

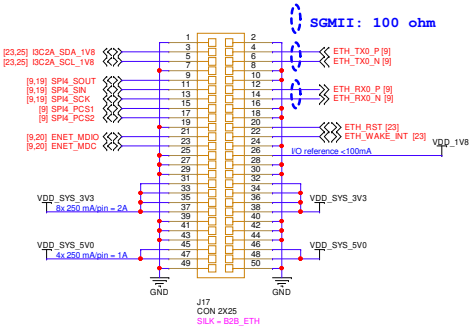
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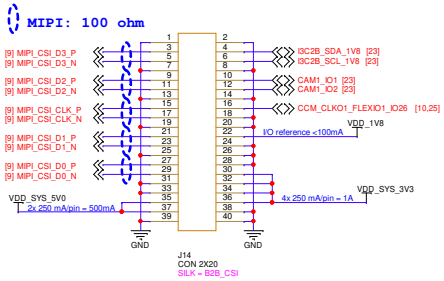
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Add-On B2B Interfaces

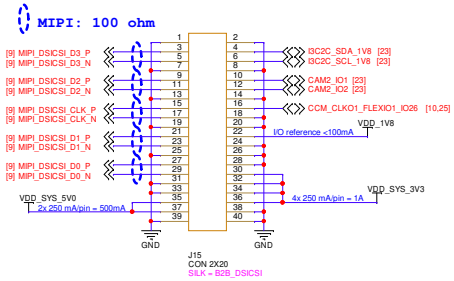
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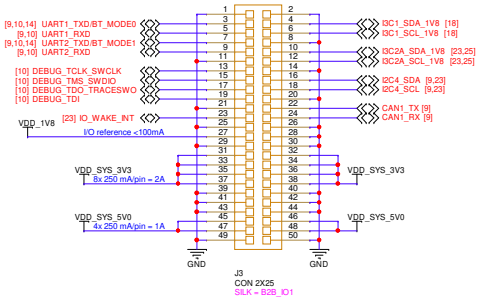
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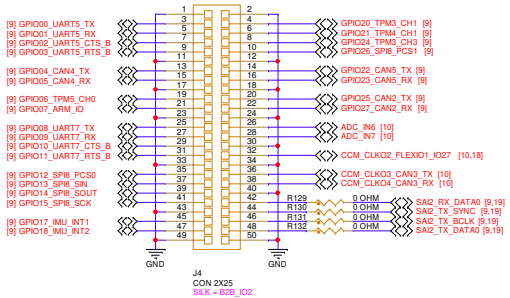


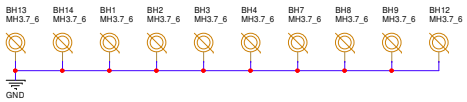
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I/O B2B







Classification: Public

Drawing Title: NAVQ95-MAIN

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